

Text-Image-GenAI Master Flashcards

STUDY PACK

Methods • Evaluation • Ethics

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Generated: 2026-05-12 15:51

Format: question and answer study cards

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[M] Methods

[E] Evaluation

[H] Ethics

Methods

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[M] Methods

Q. What are the main components of the Text-Image-GenAI pipeline?

A. Text encoder, latent diffusion model (text→image), vision-language captioner (image→text), evaluation/API.

[M] Methods

Q. What advantage do latent diffusion models provide?

A. Operate in compressed latent space for high fidelity sampling and computational efficiency compared to pixel-space diffusion.

[M] Methods

Q. What is classifier-free guidance?

A. A technique that mixes conditional and unconditional denoising to trade fidelity vs diversity via a guidance scale.

[M] Methods

Q. Name the multi-objective loss components used.

A. L_{text} (CLIP contrastive), L_{img} (diffusion reconstruction), L_{perc} (perceptual), $R(\Theta)$ (regularization).

[M] Methods

Q. What metric measures semantic alignment of image and text?

A. CLIPScore.

[M] Methods

Q. Which metric evaluates image quality?

A. FID (Fréchet Inception Distance).

[M] Methods

Q. How is round-trip evaluation performed?

A. Generate image from prompt → caption the image → compare caption to original prompt (semantic metrics).

[M] Methods

Q. Two engineering strategies to reduce compute cost?

A. Use latent-space sampling, mixed precision, caching embeddings, and smaller checkpoints or adapters.

[M] Methods

Q. Name one safety control to add in deployment.

A. Content filters, watermarking, and human review.

Evaluation

[E] Evaluation

Q. What is round-trip evaluation?

A. Generate image from prompt
→ caption image
→ compare caption to original prompt (semantic metrics).

[E] Evaluation

Q. Which metric measures image quality?

A. FID (Fréchet Inception Distance).

[E] Evaluation

Q. Which metric measures semantic alignment between image and text?

A. CLIPScore.

[E] Evaluation

Q. Name one perceptual metric used for image similarity.

A. LPIPS.

[E] Evaluation

Q. Why complement automatic metrics with human studies?

A. Automatic metrics don't fully capture realism, coherence, or bias; humans provide qualitative judgment.

Ethics

[H] Ethics

Q. Name three ethical risks of text→image models.

A. Misinformation/deepfakes, bias/stereotyping, privacy leaks.

[H] Ethics

Q. One technical mitigation for detecting synthetic images?

A. Watermarking or provenance metadata.

[H] Ethics

Q. Why is dataset curation important?

A. Reduces bias, improves provenance, and lowers privacy risks.